FORMATIVE ASSESSMENT 1 · AR / VR / XR

AR Sports Training System

An augmented-reality platform that overlays live guidance, targets and performance feedback onto real sports practice.

THE PROBLEM

Conventional sports training needs constant coach supervision, repeated physical practice and access to proper facilities. AR can overlay virtual guidance, targets, trajectories and feedback onto the real environment.

TEAM

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INTRODUCTION

The Problem & Why It Matters

Sports players often have limited access to personalized coaching, objective performance measurement, and realistic variations of training drills.

Cost

Professional coaching and specialized equipment can be expensive.

Feedback

Players mainly depend on coaches for performance evaluation.

Repetition

The same physical drills become repetitive and less engaging.

Visualization

Hard to visualize ideal trajectories, positions or movements.

TARGET USERS

- Athletes
- Sports students
- Coaches
- Fitness & training academies

WHY AR?

- Combines the real sports environment with virtual information
- Lets players interact with real equipment
- Gives real-time visual guidance
- Enables objective performance feedback

FOUNDATIONS

Understanding AR & Human Perception

Augmented Reality (AR) enhances the real-world environment by superimposing computer-generated information — 3D objects, text, targets, trajectories and instructions — onto the user's view.

HOW AR SPORTS TRAINING USES HUMAN PERCEPTION

Vision
Track targets, players and trajectories

Hearing
Audio instructions and alerts

Touch
Physical interaction with equipment

Spatial
Position, distance and movement

KEY ELEMENTS OF AN AR EXPERIENCE

Real World + Virtual Objects + Interaction + Tracking + Feedback

AR vs VR

AR	VR
Real environment stays visible	Entire environment is virtual
Virtual information is overlaid	User is immersed in virtual world
Real equipment can be used	Usually simulated equipment
Best for field-based training	Best for fully simulated training

IDEATE

From Question to Concept

HOW MIGHT WE…

“Use AR to give athletes interactive guidance and real-time performance feedback while they practice in their actual sports environment?”

POSSIBLE APPLICATIONS

1. AR Cricket Training
2. AR Football Training
3. AR Basketball Training
4. AR Tennis Training
5. AR Athletics Training

SELECTED CONCEPT

AR Multi-Sport Training System

THE SYSTEM CAN PROVIDE:

- ✓ Virtual targets
- ✓ Movement guidance
- ✓ Ball trajectory visualization
- ✓ Timing indicators
- ✓ Training drills
- ✓ Real-time performance scoring

VIRTUAL WORLD

Representing the Virtual World

The virtual training environment is made of digital elements placed within the real sports environment, expressed through three channels.

◈ Visual

- 3D virtual targets
- Ball trajectory
- Player position markers
- Movement arrows
- Distance indicators
- Training zones
- Scoreboards

♪ Aural

- Voice instructions
- Beep / timing signals
- Success / failure sounds
- Performance announcements

✋ Haptic

- Controller / device vibration
- Wearable feedback
- Equipment-based feedback

VIRTUAL WORLD FLOW

Objects > Position > Movement > Interaction > Feedback

SYSTEM DESIGN

Proposed AR Sports Training System



CORE COMPONENTS

- Camera
- Motion / position tracking
- AR display
- Training scenario engine
- Sports database
- Performance analyzer
- Feedback system

AR EXPERIENCE

Key Elements of the AR Training Experience

1

Tracking

Detects the player's position, environment and movement.

2

Registration

Places virtual objects correctly within the real environment.

3

Visualization

Displays targets, trajectories and instructions.

4

Interaction

Lets the player respond to virtual targets and instructions.

5

Feedback

Gives immediate information about performance.

6

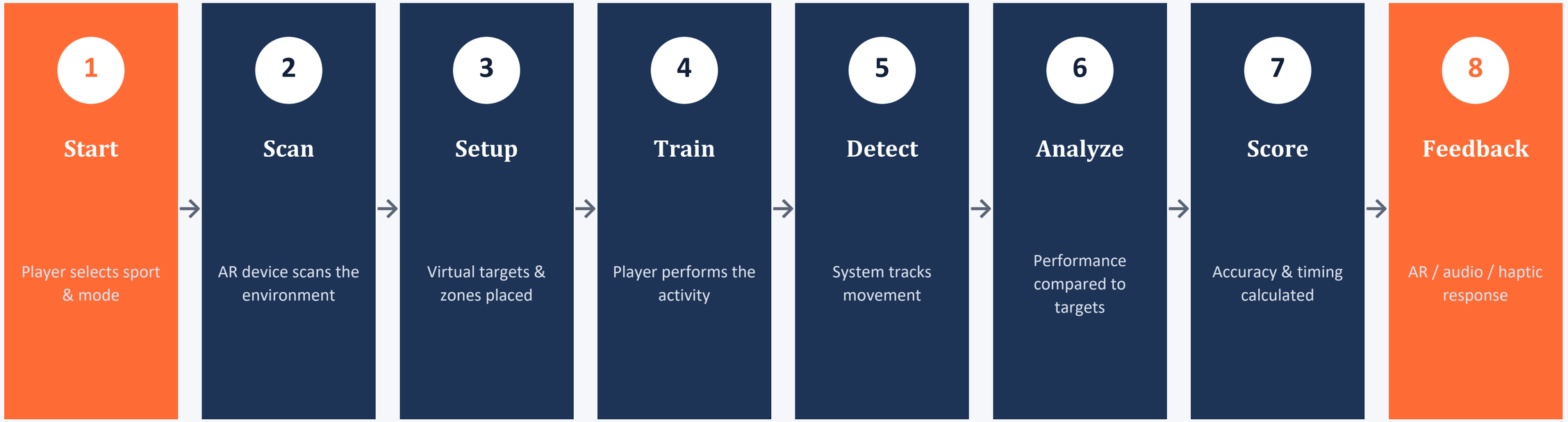
Immersion & Presence

How strongly the tech engages the senses, and how involved the player feels.

Immersion: how strongly the technology surrounds and engages the user's senses. **Presence:** the user's psychological feeling of being involved in the training environment.

WORKFLOW

A Training Session, Step by Step



A full loop — scan, train, detect, and respond — repeats every session, adapting drills as the player's data builds up over time.

IMPACT

Applications, Benefits & Future Scope

APPLICATIONS

- Cricket batting & bowling practice
- Football shooting & passing
- Basketball shooting
- Tennis stroke training
- Athletics movement training
- Sports rehab & skill development

BENEFITS

Personalized Training

Adjusts to player performance.

Real-Time Feedback

Players immediately see mistakes.

Interactive Learning

Makes training more engaging.

Performance Tracking

Records accuracy, timing, progress.

FUTURE SCOPE

- AR smart glasses
- AI-based coaching
- Cloud performance analytics
- Haptic sports equipment
- Multi-player AR training
- Biometric & motion tracking

From drills to data-driven training

AR turns every practice session into measurable, personalized improvement.

CONCLUSION

Train Smarter. Perform Better.

The AR Sports Training System combines the real sports environment with computer-generated visual, aural and haptic information to create an interactive training experience — helping athletes improve accuracy, timing, movement and decision-making through real-time guidance.

By combining tracking, virtual-world representation, interaction, immersion and presence, AR can make sports training more personalized, engaging and measurable.

KEY CONCEPT

Real Sports Environment + AR Virtual Information + Player Interaction + Performance Feedback

"Train smarter. Visualize better. Perform better."